Research and developments on the eastern Bering Sea walleye pollock stock assessment

September 2025, Plan Team Draft

Jim Ianelli, Carey McGilliard, Sophia Wassermann

2025-08-27 22:19

Executive Summary

The work presented below was based on the same data sets used in the 2024 assessment. The first task was to update the model runs to include the 2023 data. The model runs were updated and the results are presented below. The model runs were then used to compare the FT-NIR data with the traditional data sources (BTS and ATS). The results of these comparisons are presented below. The model fits to the data are presented in the tables at the end of the document. The first section steers towards the R library used for processing model results for EBS pollock. The second section presents the comparisons of the FT-NIR data with the traditional data sources. The third section compares results of different software platforms (e.g., SS3, CEA, and AMAK) for the assessment. The fourth section shows results comparing newer model-based estimates from survey data.

Response to 2025 CIE Review of EBS Walleye Pollock Assessment

This section outlines the response to the findings and recommendations of the 2025 Center for Independent Experts (CIE) review of the Eastern Bering Sea (EBS) walleye pollock stock assessment. The final peer-reviewed reports from the three reviewers are provided at this link https://www.st.nmfs.noaa.gov/science-quality-assurance/cie-peer-reviews/cie-review-2025 . These documents summarize their findings and recommendations in detail.

Since the review:

- The operational ADMB-based model has been fully ported to **RTMB**, with verification that outputs match the prior version.
- Initial steps are underway to address **Russian catch** uncertainty through scenario modeling based on available data.

---

1. Model Platform and Estimation Framework

The full ADMB model has been implemented in RTMB, enabling: - Random walk selectivity and recruitment deviations - Improved error propagation via marginal likelihood - Mohn's rho, retrospective peels, and full simulation integration

Status: Complete
See attached illustration chunk for output comparison (add figures/tables here).

---

2. Russian Removals and Stock Exchange

Russian catch has not historically been included in the assessment. All three reviewers noted this omission and recommended: - Conducting **sensitivity runs** - Quantifying the **scale and direction of transboundary movement** - Developing **collaborative or joint modeling strategies**

Status: In progress
Ongoing work involves sensitivity runs with assumed Russian removals of 20–45% of the EBS US catch, guided by UCSL (2024) and reviewer recommendations.

---

3. Process Uncertainty, Selectivity, and Growth

Key issues raised: - Random walk selectivity penalties are currently fixed. Reviewers suggested estimating process variances via marginal likelihood (Anders). - Weight-at-age models could be expanded to include cohort and temperature effects (Yan). - Estimation of time-varying natural mortality should be explored cautiously (Yan).

Status: Partially addressed — EDF and penalty variance exploration will proceed after Russian catch integration.

---

4. Data Weighting and Compositional Likelihoods

Matt Cieri strongly recommended testing a **Dirichlet-multinomial (DM)** approach via Laplace approximation to better reflect age composition variance.

Status: Under review — current bootstrap-based effective N retained; DM implementation deferred until after RTMB stabilization.

Table 1: CIE 2025 Recommendations and Response Summary

Theme	Lead Reviewer	Recommendation	Status
RTMB transition	All	Port model from ADMB to RTMB	Com
Russian catch	All	Incorporate or test Russian removals	In pr
Selectivity complexity	Nielsen	Estimate process variances via marginal likelihood	Plan
Natural mortality	Jiao	Explore M variation and CEATTLE integration	Conc
Growth & WAA	Jiao	Model cohort and temperature effects on WAA	Defe
Data weighting	Cieri	Test Dirichlet-multinomial likelihoods	Defe
Retrospective diagnostics	Nielsen	Use Mohn's rho and integrate peels	Com

5. Retrospective Performance

Mohn's rho was computed and incorporated into RTMB outputs. Diagnostic plots are available for: - Spawning biomass (SSB) - Recruitment (R) - Fishing mortality (F)

Status: Complete

---

6. Summary of Reviewer Recommendations and Status

```
tribble(
  ~Theme, ~`Lead Reviewer`, ~Recommendation, ~Status,
  "RTMB transition", "All", "Port model from ADMB to RTMB", "  Completed",
  "Russian catch", "All", "Incorporate or test Russian removals", "  In progress",
  "Selectivity complexity", "Nielsen", "Estimate process variances via marginal likelihood",
  "Natural mortality", "Jiao", "Explore M variation and CEATTLE integration", "  Conceptual",
  "Growth & WAA", "Jiao", "Model cohort and temperature effects on WAA", "  Deferred",
  "Data weighting", "Cieri", "Test Dirichlet-multinomial likelihoods", "  Deferred",
  "Retrospective diagnostics", "Nielsen", "Use Mohn's rho and integrate peels", "  Completed"
) %>%
  gt::gt()
```

ADMB → RTMB Code

subtitle: "Tracking translation details while bridging ADMB to RTMB"

This standalone appendix tracks items to verify or refactor as the model is bridged from **ADMB** to **RTMB**. Each entry records the ADMB behavior, intended RTMB behavior, priority, and links/tests.

Model implementation

The RTMB model code is implemented in the `R/` directory, with the main entry point in `R/Rpm.R`.

Comparison of ADMB and RTMB by components

The initial steps to bridge ADMB to RTMB involve identifying key components and their behaviors. The following table summarizes the components, their ADMB behavior, intended RTMB behavior, and current status (Table 1).

After some initial testing we were able to take the parameter estimates from the ADMB model and show that the gradients were very close to zero and the negative log-likelihood values matched.

outer mgc: 6.026919e-06 outer mgc: 6.026919e-06

Reparameterizing dev-vectors

The ADMB code has lots of cases where say y_i is a function of a parameter θ and a deviation ϵ_i such that $y_i = \theta + \epsilon_i$. In ADMB, the deviations are estimated as a vector of parameters with a constraint to have mean zero., In RTMB, it seems best to reparameterize them as a simple vector of parameters (say as θ_i) and then do the likelihood parts as deviations from the mean.

So far we've done this for the following parameters:

- `log_avgrec` (average recruitment)

Table 1: Comparison of the rtmb code (`RPM`) and the ADMB code results.

Opening /Users/jim/__mymods/noaa-afsc/EBS__pollock/runs/data/wtage2024.dat
[

Comparison of RTMB and ADMB Outputs
Tolerance = 1e-05] Comparison of RTMB and ADMB Outputs
Tolerance = 1e-05

Variable	Equal (tol)	Length	Max |Abs diff|	Max |% diff|	Correlation
N	TRUE	915	0.049666	0.000439	1.000000
Z	TRUE	915	0.000000	0.000151	1.000000
F	TRUE	915	0.000000	0.000491	1.000000
M	TRUE	915	0.000000	0.000000	1.000000
S	TRUE	915	0.000000	0.000120	1.000000
pred__catch	TRUE	61	0.004832	0.000391	1.000000
obs__catch	TRUE	61	0.005000	0.000458	1.000000
SSB	TRUE	61	0.004954	0.000428	1.000000
phizero	TRUE	1	0.000000	0.000189	NA
Bzero	TRUE	1	0.000761	0.000013	NA
steepness	TRUE	1	0.000000	0.000054	NA
pred__cpue	TRUE	12	0.004774	0.000137	1.000000
pred__avo	TRUE	18	0.000005	0.000314	1.000000
eb__bts	TRUE	42	0.022933	0.000212	1.000000
eb__ats	TRUE	19	0.004530	0.000149	1.000000
sel__fsh	TRUE	915	0.000005	0.000467	1.000000
sel__bts	TRUE	645	0.000000	0.000398	1.000000
sel__ats	TRUE	465	0.000005	0.000491	1.000000
sam__fsh	TRUE	60	0.000499	0.000134	1.000000
sam__bts	TRUE	42	0.000000	0.000000	1.000000
sam__ats	TRUE	19	0.000000	0.000000	1.000000
phat__fsh	TRUE	900	0.000000	0.000476	1.000000
phat__bts	TRUE	630	0.000000	0.000468	1.000000
phat__ats	TRUE	285	0.004637	0.000442	1.000000
age__like	TRUE	3	0.001383	0.000212	1.000000
age__like__offset	TRUE	3	0.039496	0.000326	1.000000
cat__like	TRUE	1	0.000002	0.000078	NA
bts__like	TRUE	1	0.000004	0.000011	NA
ats__like	TRUE	1	0.000003	0.000027	NA
ats__age1__like	TRUE	1	0.000039	0.000346	NA
cpue__like	TRUE	1	0.000002	0.000174	NA
avo__like	TRUE	1	0.000002	0.000020	NA
avgsel__like	TRUE	1	0.000000	0.000235	NA
wt__nll	TRUE	4	0.004116	0.000190	1.000000
wt__like	TRUE	1	0.004805	0.000076	NA
rec__like	TRUE	7	0.000047	0.000460	1.000000
Fpen__like	TRUE	1	0.000004	0.000043	NA
sel__like	TRUE	3	0.000023	0.000090	1.000000
sel__like__dev	TRUE	3	0.000338	0.000287	1.000000
Priors	TRUE	4	0.000006	0.000029	1.000000
tot__like	TRUE	1	0.002266	0.000028	NA

Table 2: Top 20 parameters with largest gradients (absolute value) from RTMB using ADMB parameter estimates.

Parameter	gradient
sel_slp_bts_dev	-6.026919e-06
sel_devs_ats	-5.780988e-06
sel_devs_ats	-5.442764e-06
sel_devs_ats	-5.254151e-06
sel_devs_ats	5.210384e-06
sel_devs_ats	5.065771e-06
sel_devs_ats	5.020803e-06
sel_devs_ats	-4.987770e-06
sel_devs_ats	4.963619e-06
sel_devs_ats	4.907284e-06
sel_devs_ats	4.413068e-06
sel_devs_ats	4.319045e-06
sel_devs_ats	-4.250591e-06
sel_devs_ats	4.144523e-06
sel_devs_ats	-4.106040e-06
sel_devs_ats	-3.973683e-06
sel_devs_fsh	-3.950742e-06
sel_devs_ats	-3.928872e-06
sel_devs_ats	3.832048e-06
sel_devs_fsh	3.822259e-06

- `log_avg_F` (average fishing mortality)
- `log_slp_bts` (slope of the selectivity for the bottom trawl)
- `log_a50_bts` (age-50% selected for the bottom trawl)

The results from feeding the MLE parameter estimates from ADMB into the RTMB code gave exactly the same estimates of spawning stock biomass (SSB) and age 1 recruits (Figure 1; Figure 2).

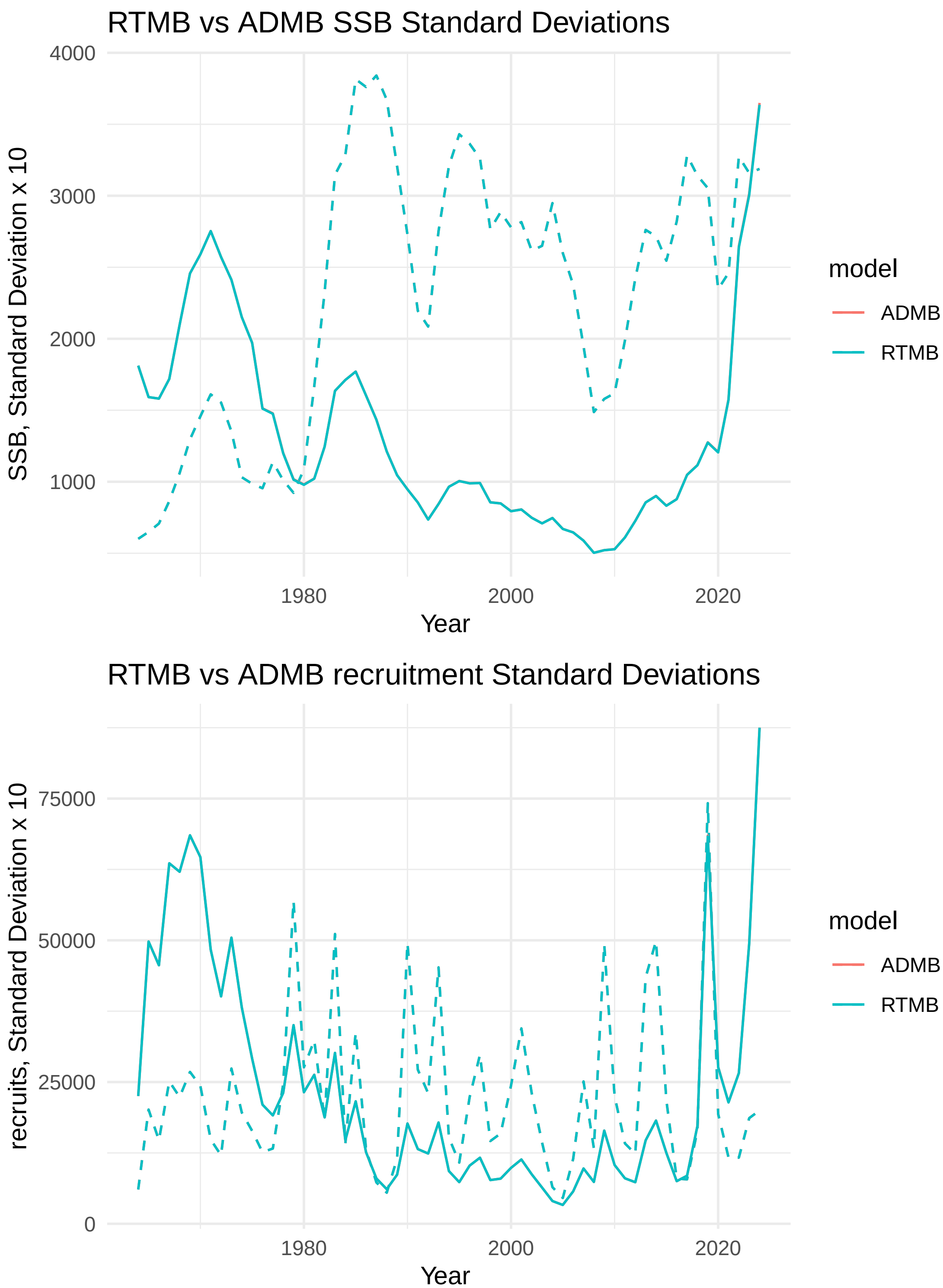

Figure 1: SSB and age-1 recruits from RTMB and ADMB, dashed lines are standard deviations (x10). Both ADMB and RTMB results are overlain.

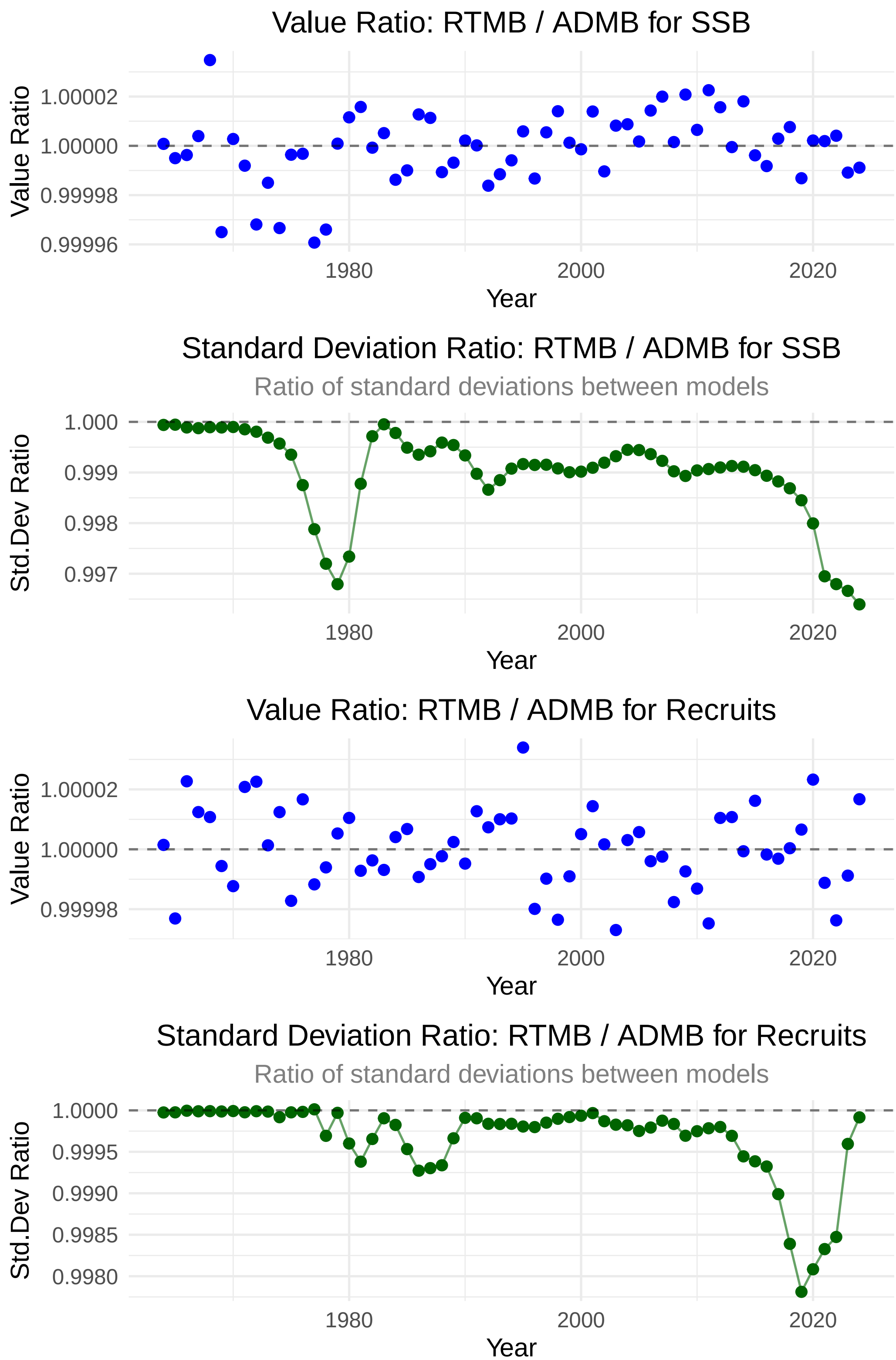

Figure 2: Relative differences between ADMB and RTMB estimates of SSB and age-1 recruits and asymptotic standard deviation approximations.

Considerations for future assessments regarding neighboring Russian federation fisheries

As this year's assessment moves forward, particular attention is warranted on how Russian fisheries and assessments are considered alongside our own results. Pollock in the eastern and western Bering Sea are generally recognized as a single biological stock, with seasonal exchange across the U.S.–Russia boundary documented by acoustic studies. As noted in the CIE reviews, Russian removals are substantial with recent estimates representing on the order of 20–45% of the U.S. catch in some years.
While examined in past assessments, the Russian catch magnitude should be re-evaluated to examine how they might alter management advice. As a step in beginning to make a more detailed evaluation, we compared the Russian stock assessment results presented as part of their Marine Stewardship Council (MSC) review panel with our model results (through 2024). The following sections compare outcomes from the assessments presented to

Comparison of Russian and USA age-structured assessment results

```
=== CORRELATION ANALYSIS ===
```

Pollock 99% of catch; seabird & marine mammal mortality very low (Artyukhin, Korobov, and Gluschenko 2023; Burdin 2021). (Japp and Sharov 2023, 2024; Bulatov and Vasilev 2023).

[1] "mean ratio (USA/Russian): 2.87" [1] "median ratio (USA/Russian): 2.7"

```
=== DETAILED STATISTICS ===
```

```
Summary statistics by country and age:
```

```
# A tibble: 20 x 8
   Country   Age    Mean  Median      SD    CV    Min    Max
   <chr>   <dbl>   <dbl>   <dbl>   <dbl> <dbl>  <dbl>  <dbl>
```

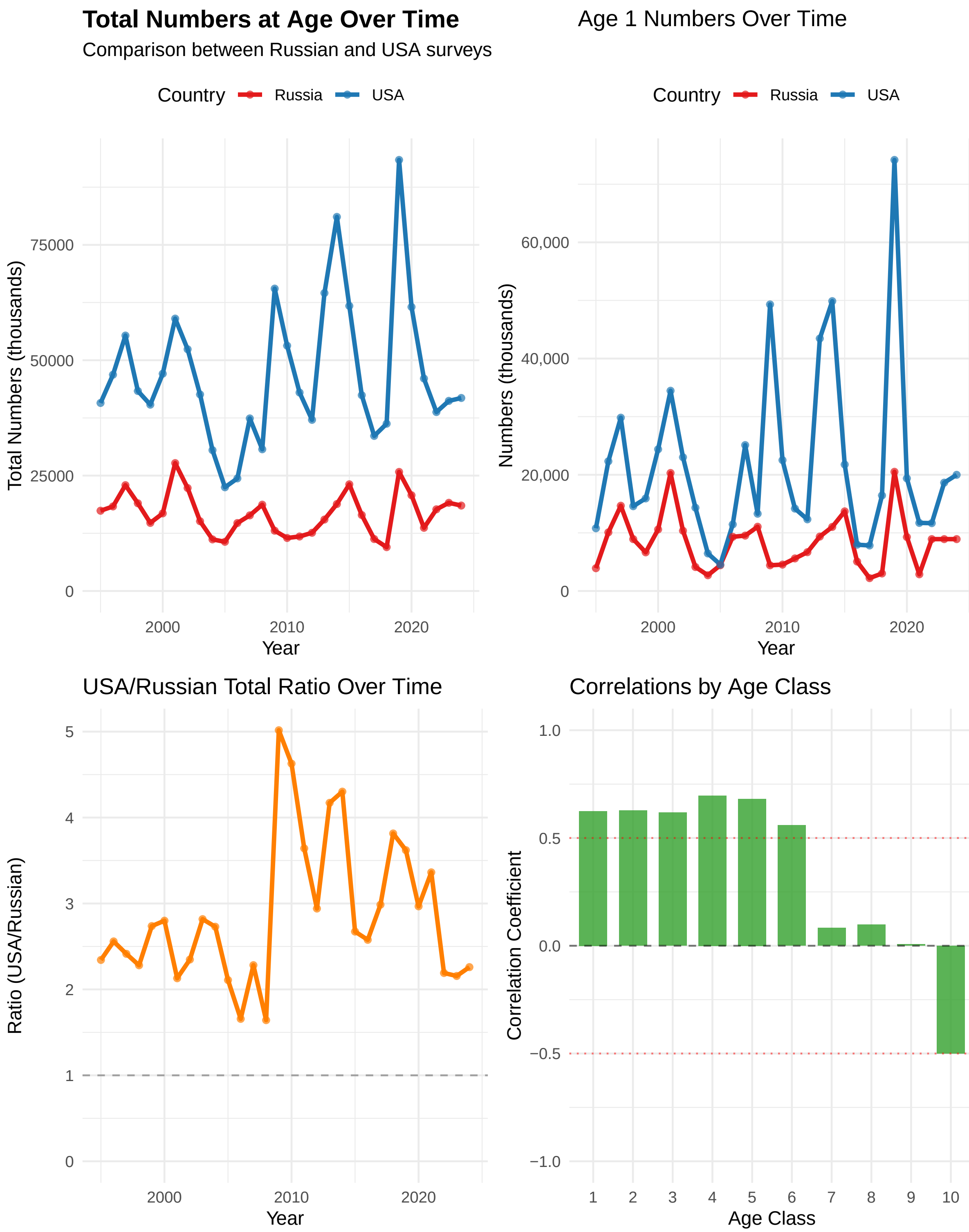

Figure 3: Various comparisons between Russian and USA age-structured survey results.

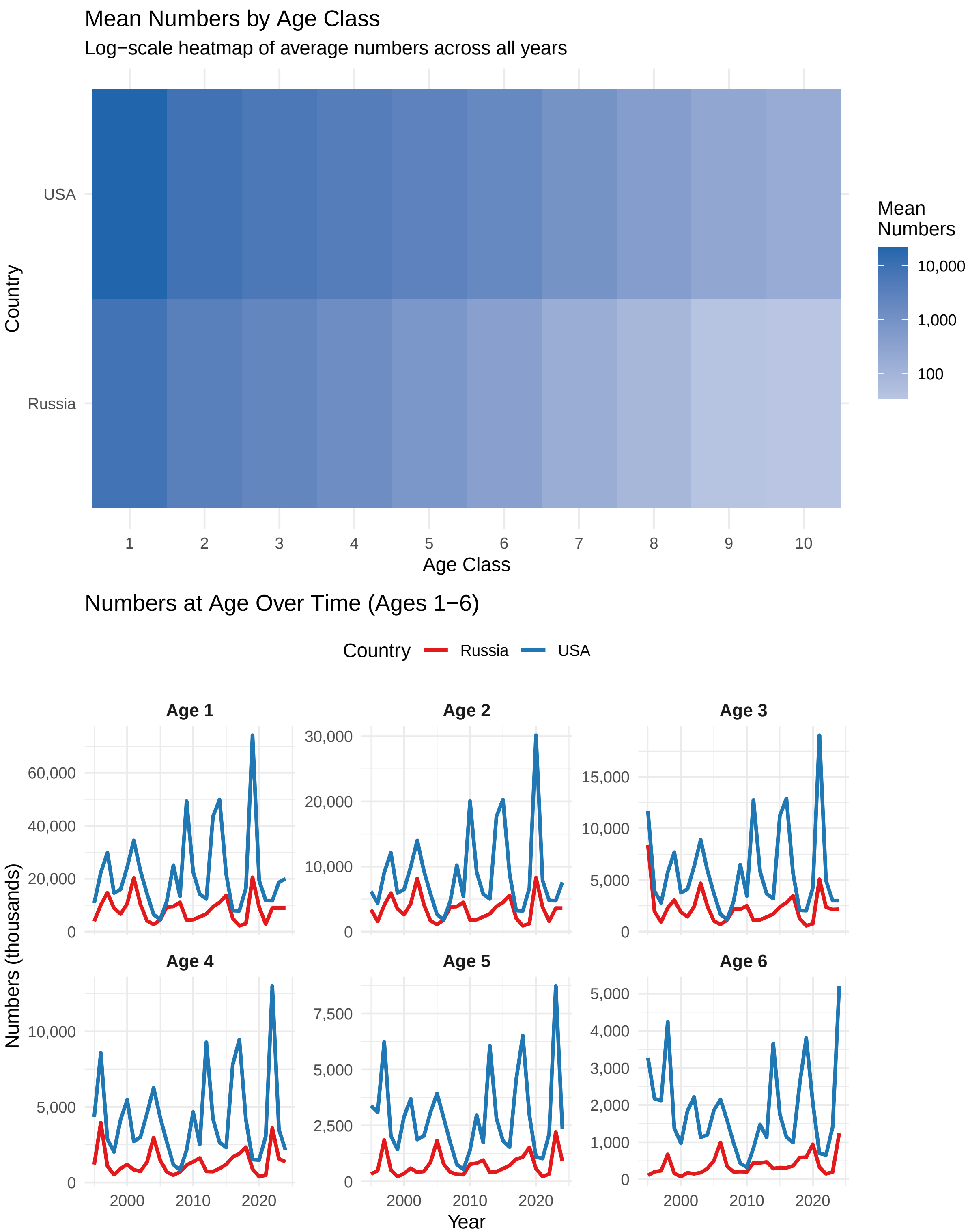

Figure 4: Various comparisons between Russian and USA age-structured survey results.

Table 1: Correlations between Russian and USA age classes, 1995-2024.

[

Correlations between Russian and USA Age Classes] Correlations between Russian and USA Age Classes

Age	Correlation
1	0.625
2	0.627
3	0.618
4	0.697
5	0.682
6	0.560
7	0.083
8	0.098
9	0.007
10	-0.500

```
 1 Russia      1  8391.   8925    4647.  0.554 2213.  20509.
 2 Russia      2  3400.   3562.   1860   0.547  900.   8308.
 3 Russia      3  2254.   2154.   1574.  0.699  565.   8406.
 4 Russia      4  1338.   1169.    875   0.654  387.   3966.
 5 Russia      5   743     575.    511.  0.688  216.   2214.
 6 Russia      6   382.    302.    278   0.728   74.5  1242.
 7 Russia      7   168.    138.    127.  0.757   15.3   550.
 8 Russia      8    89.3    65.2    81.9 0.917    0.8   367.
 9 Russia      9    38.7    31.2    37.3 0.964    0.4   178.
10 Russia     10    35.1    16.8    45.4 1.29     0.2   164.
11 USA         1 21725.  17539.  15273.  0.703 4546.  74180.
12 USA         2  8768.   6580.   6227.  0.71  1848.  30158.
13 USA         3  5797.   4186    4098.  0.707 1175.  19030.
14 USA         4  4232.   3283.   2852.  0.674  831.  12994.
15 USA         5  2912.   2594.   1889.  0.648  543.   8729.
16 USA         6  1842.   1535.   1174.  0.637  325    5198.
17 USA         7   935.    809.    565.  0.604  185.   2560.
18 USA         8   488     375.    317.  0.649   88.1  1532
19 USA         9   260.    204.    178.  0.685   41.8   861.
20 USA        10   188     176.    103.  0.547   37.8   476.


Coefficient of Variation by age:
```

```
# A tibble: 10 x 3
     Age Russia_CV USA_CV
   <dbl>     <dbl>  <dbl>
 1     1     0.554  0.703
 2     2     0.547  0.71
 3     3     0.699  0.707
 4     4     0.654  0.674
 5     5     0.688  0.648
 6     6     0.728  0.637
 7     7     0.757  0.604
 8     8     0.917  0.649
 9     9     0.964  0.685
10    10     1.29   0.547
```

```
Age classes with correlation > 0.5: Age1, Age2, Age3, Age4, Age5, Age6
```

```
Age classes with correlation < 0.2: Age7, Age8, Age9, Age10
```

Results comparing Russian and USA age-structured assessments .

Other items

The CIE noted in particular that some documentation of aspects of the ADMB code was incomplete. During the meetings, we began to more properly specify how the selectivity penalties were being calculated and implementated.

First-difference selectivity penalty (temporal)

We need the penalty used in ADMB where the **first year is compared to zero** (not to a previous estimated year). For a matrix of log-selectivities ($\log s_{y,a}$) (years($y = 1, \ldots, T$), ages ($a = 1, \ldots, A$), the penalty is:

$$\mathcal{P}_{\text{FD,time}} = w \sum_{a=1}^{A} \Big[\big(\log s_{1,a} - 0 \big)^2 + \sum_{y=2}^{T} \big(\log s_{y,a} - \log s_{y-1,a} \big)^2 \Big].$$

RTMB/R implementation

```r
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# w: scalar or length-A vector of weights
fd_penalty_time <- function(log_sel, w = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  if (length(w) == 1) w <- rep(w, A)
  stopifnot(length(w) == A)

  diffs <- rbind(log_sel[1, , drop = FALSE] - 0,   # year 1 vs 0
                 apply(log_sel, 2, diff))          # y>=2 diffs
  sum(colSums((sqrt(w) * diffs)^2))
}
```

Sanity check

```r
set.seed(42)
T <- 5; A <- 3
L <- matrix(rnorm(T*A, sd=0.2), T, A); w <- c(1, 2, 0.5)

fd_loop <- function(L, w){
  acc <- 0
  for (a in seq_len(ncol(L))) {
    acc <- acc + w[a]*(L[1,a]-0)^2
    for (y in 2:nrow(L)) acc <- acc + w[a]*(L[y,a]-L[y-1,a])^2
  }
  acc
}

all.equal(fd_penalty_time(L, w), fd_loop(L, w))
```

```
[1] "Mean relative difference: 0.01953004"
```

First-difference selectivity penalty (age)

For within-year smoothing over age with a **zero anchor at the minimum age**:

$$\mathcal{P}_{\mathrm{FD,age}} = w \sum_{y=1}^{T} \Big[\big(\log s_{y,a_{\min}} - 0 \big)^2 + \sum_{a=a_{\min}+1}^{A} \big(\log s_{y,a} - \log s_{y,a-1} \big)^2 \Big].$$

RTMB/R implementation

```
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# a_min: column index of minimum age within 'log_sel' (default = 1)
fd_penalty_age <- function(log_sel, w = 1, a_min = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  stopifnot(a_min >= 1, a_min <= A)
  if (length(w) == 1) w <- rep(w, T)
  stopifnot(length(w) == T)

  # Build diffs across age with anchor at a_min (assumes a_min=1; if not, set zeros below a_
  diffs <- matrix(0, nrow = T, ncol = A)
  diffs[, a_min] <- log_sel[, a_min] - 0
  if (a_min < A) {
    for (a in (a_min+1):A) {
      diffs[, a] <- log_sel[, a] - log_sel[, a-1]
    }
  }
  sum(rowSums((sqrt(w) * diffs)^2))
}
```

Reference: survey lognormal scaling (consistency check)

When porting lognormal survey likelihoods, confirm whether ADMB's objective includes the normalizing constant and whether ( ) is on the **log** scale:

$\ell(\sigma) = -\sum_i \frac{(\log O_i - \log E_i)^2}{2\sigma^2} - n \log \sigma - \frac{n}{2} \log(2\pi)$.

```
lognorm_nll <- function(obs, exp, sigma_log, include_const = TRUE) {
  r <- log(pmax(obs, .Machine$double.eps)) - log(pmax(exp, .Machine$double.eps))
  n <- length(r)
  nll <- sum(r^2) / (2 * sigma_log^2)
  if (include_const) nll <- nll + n*log(sigma_log) + n*0.5*log(2*pi)
  nll
}
```

R Library for assessment compilations (ebswp)

The `ebswp` library is a collection of functions for compiling and analyzing results from the stock assessment models for the eastern Bering Sea walleye pollock (Ianelli (2025)). The library is designed to work with the ADMB code and depends on a consistent format for the report files generated. The link in the reference provides access to some notes on applications as vignettes to the package (e.g., . here).

Comparing FT-NIR compositions with traditional microscope age estimations

The fish age and growth lab at the AFSC have endeavored to expand on new technologies (Helser et al. (2019)).

FT-NIRS data consistency comparisons

Examining the abundance-at-age (for design-based estimates of the bottom-trawl age compositions) we show that in general, the TMA estimates show a high level of consistency by cohort (Figure 5). For these same data, when applied to the FTNIRs age-estimation method, the consistency was lower (Figure 6). The extent that this is a concern is unknown.

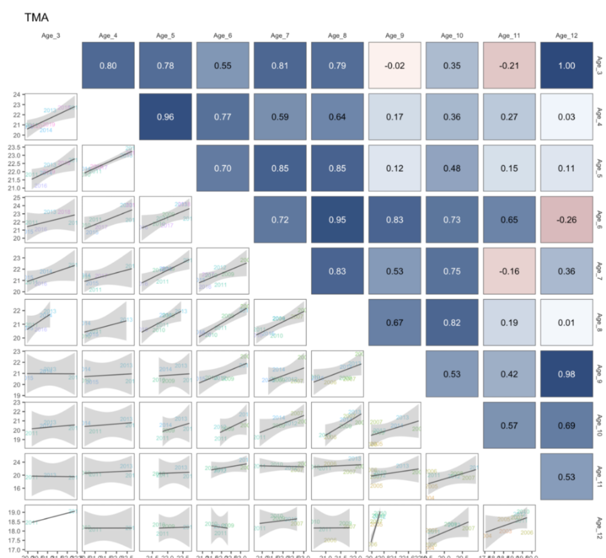

Figure 5: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for traditional microscope age estimates. The cohorts are shown in the labels.

When preparing the data for the assessment, we plotted out the proportions-at-age for the TMA data and noted the differences in the estimates using the FTNIRs age estimates (Figure 7,

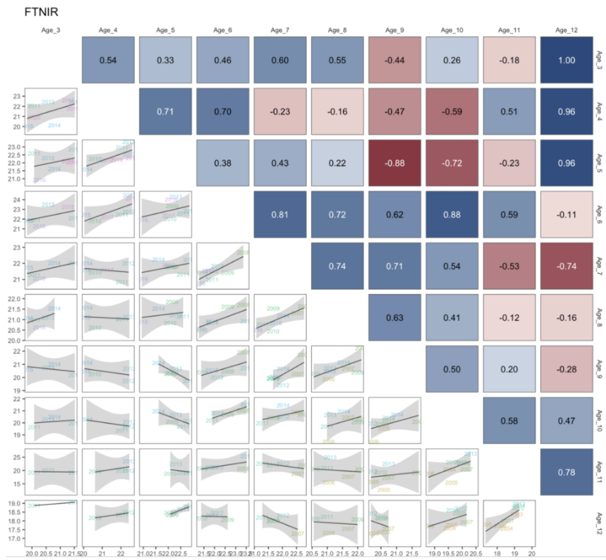

Figure 6: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for FTNIRs age estimates. The cohorts are shown in the labels.

Figure 8, and Figure 9). Other preparations involved estimating the age-determination errors for each method following that of Punt et al. (2008). For the FTNIRs data, we optionally estimated separate age-estimation error matrices by gear type. The appearance of the fleet-aggregated and fleet-specific input data files is shown in Figure 10.

For the assessment model, the alternatives are shown based on the coverage of data and availability among types of ageing is shown in Figure 11. These figures are intended to show what data are available for the model testing and the different ways of treating age determination error matrices.

Comparisons of fit to FT-NIRS models

The layout of model runs comparing the traditional microscope ages (TMA) with the FT-NIRS ages is shown in Table 1. The models are run with the FT-NIRS data with different age-error matrix specficiations. The model fits to the data are shown in Table 2. The table shows the goodness-of-fit

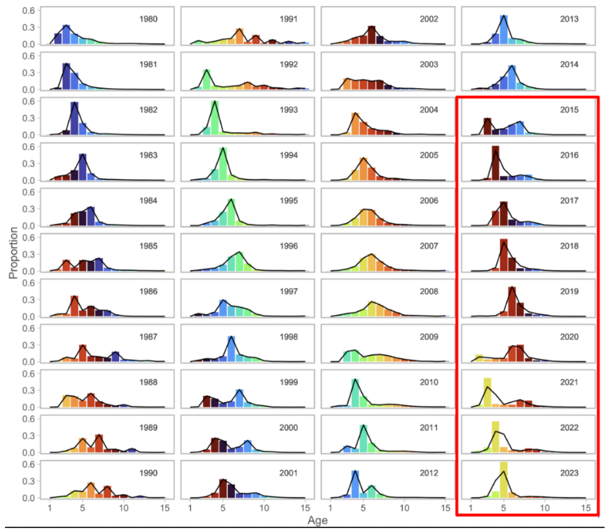

Figure 7: Proportions at age for recent years for the fishery data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

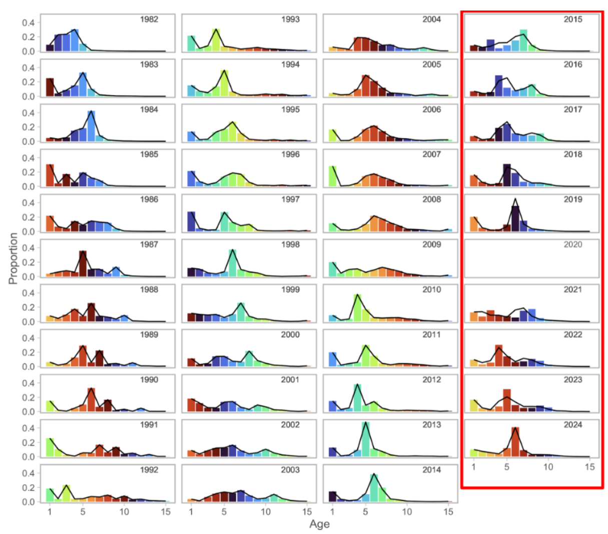

Figure 8: Proportions at age for recent years for the bottom-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

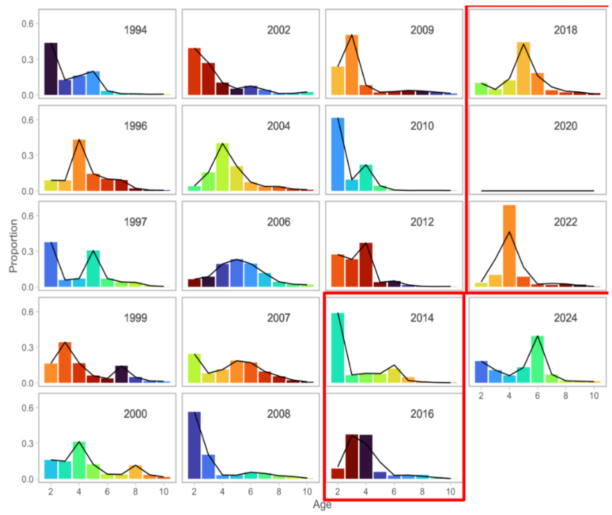

Figure 9: Proportions at age for recent years for the acoustic-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

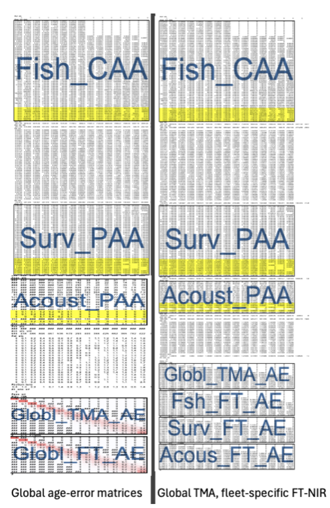

Figure 10: Schematic of input datafile showing the breakout of the fleet-specific FTNIRs age-error matrices (right panel) compared to that where the global FTNIRs age-error matrices are used (left panel).

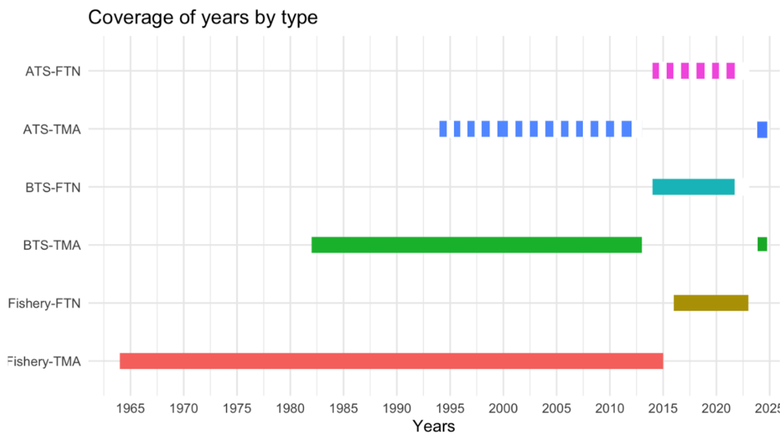

Figure 11: Data extent by age composition for the models with FTNIRs age estimates. The data types are for the Fishery, bottom trawl survey (BTS), and acoustic trawl survey (ATS) and the years of coverage are shown in the x-axis.

Table 1: Configuration of different model options testing the impact of different treatment of the age composition data and ageing error.

	Model description	
Name	Age type	Age-error
Base	TMA	Global TMA
FT-NIR 1	TMA + FT-NIR	Global TMA, global FT-NIR
FT-NIR 2	TMA + FT-NIR	Global TMA, disaggregated FT-NIR

Table 2: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	Base	FT-NIR fleets aggregated	FT-NIR fleet-specific
RMSE BTS	0.150	0.151	0.152
RMSE ATS	0.167	0.163	0.158
RMSE AVO	0.216	0.217	0.215
RMSE CPUE	0.097	0.097	0.097
SDNR BTS	0.890	0.900	0.880
SDNR ATS	0.930	0.910	0.880
SDNR AVO	0.950	0.960	0.940
Eff. N Fishery	358	336	381
Eff. N BTS	159	148	153
Eff. N ATS	233	221	233
Catch NLL	3	4	3
BTS NLL	24	24	23
ATS NLL	8	7	7
ATS Age1 NLL	7	8	9
AVO NLL	8	8	8
Fish Age NLL	417	559	614
BTS Age NLL	286	330	313
ATS Age NLL	43	42	41
NLL selectivity	213	216	215
NLL Priors	20	20	20
Data NLL	807	995	1,032
Total NLL	1110.900	1298.930	1332.810

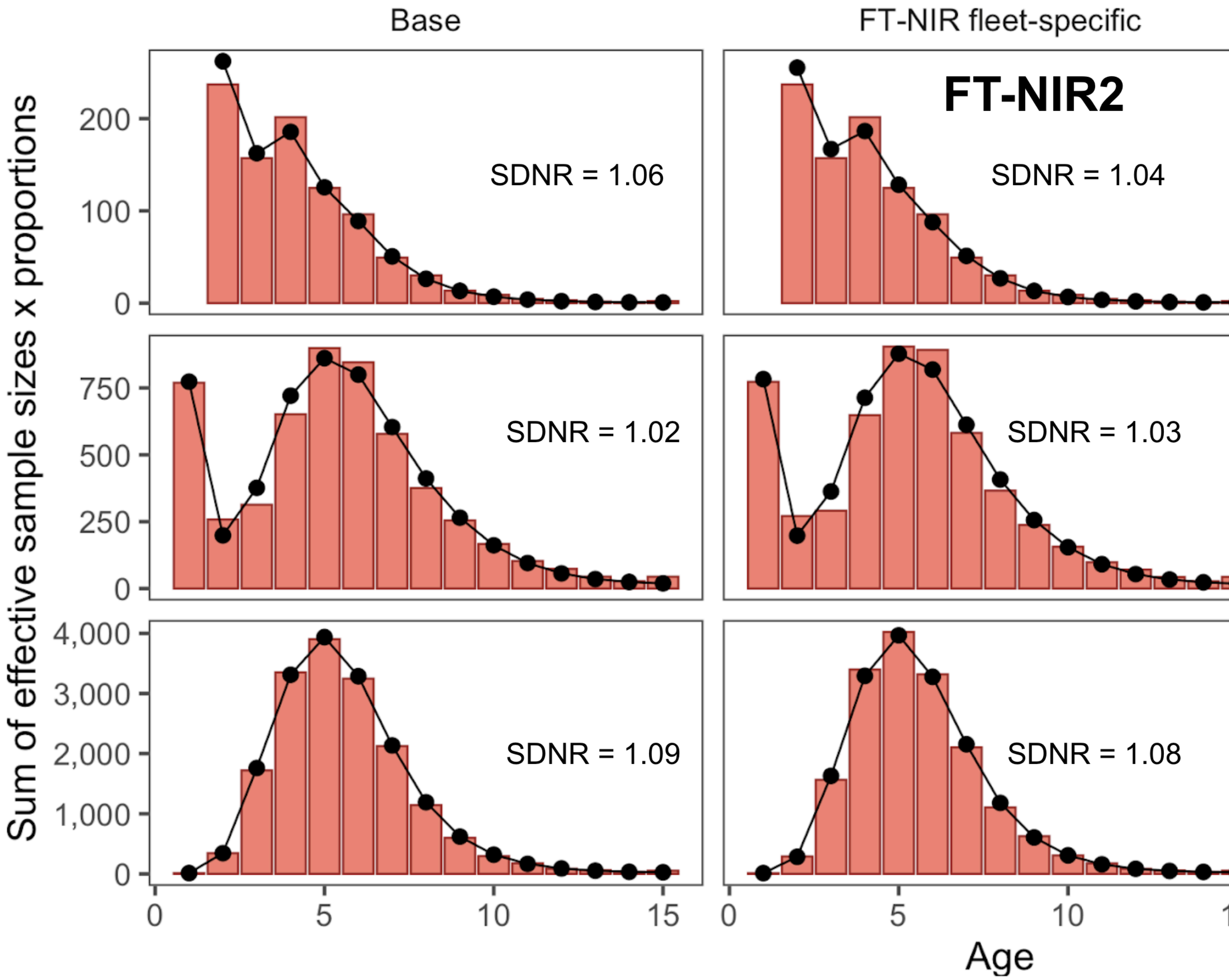

Figure 12: One-step ahead residuals for different model specifications for the fishery composition data

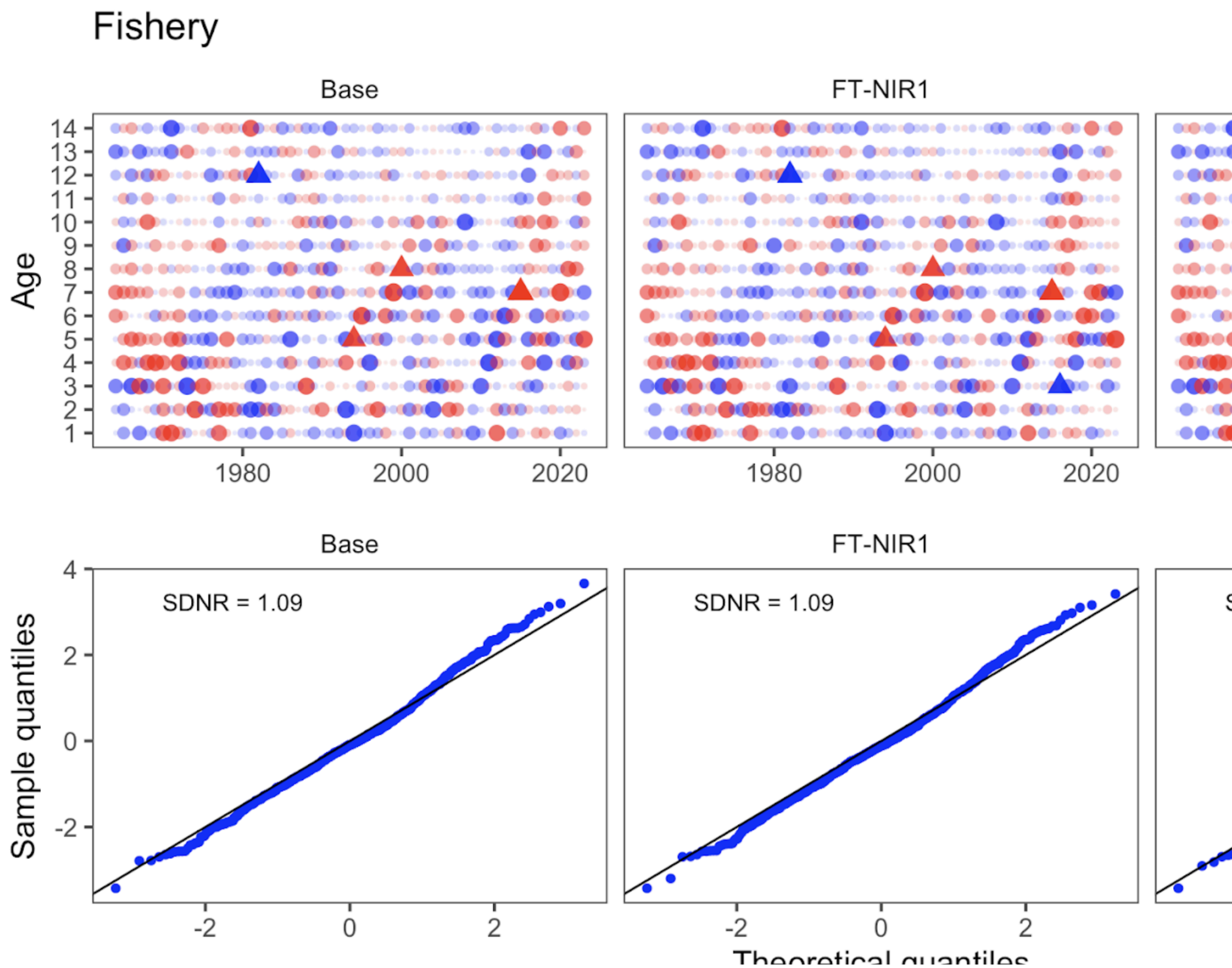

Figure 13: One-step ahead residuals for different model specifications for the fishery composition data

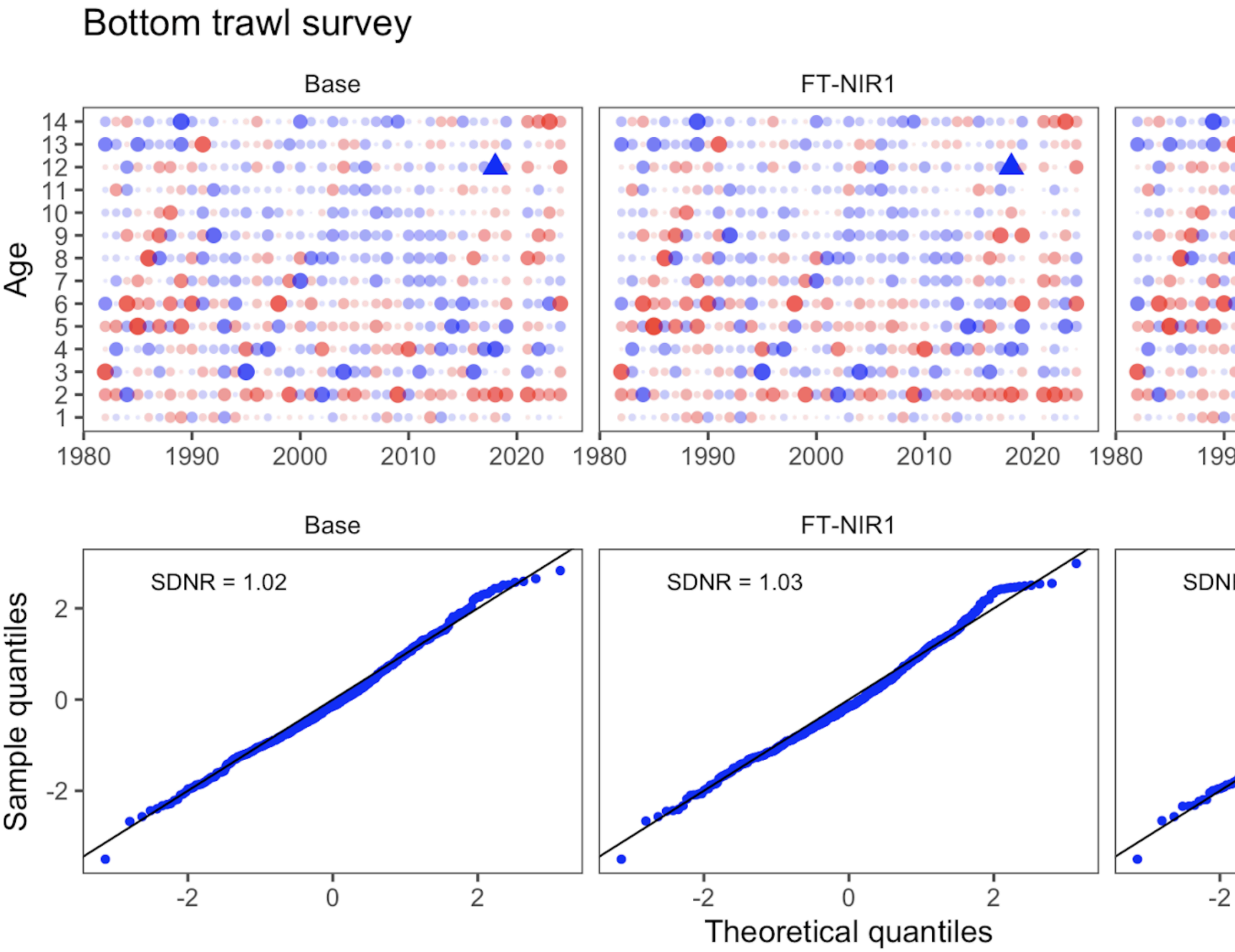

Figure 14: One-step ahead residuals for different model specifications for the bottom-trawl survey composition data

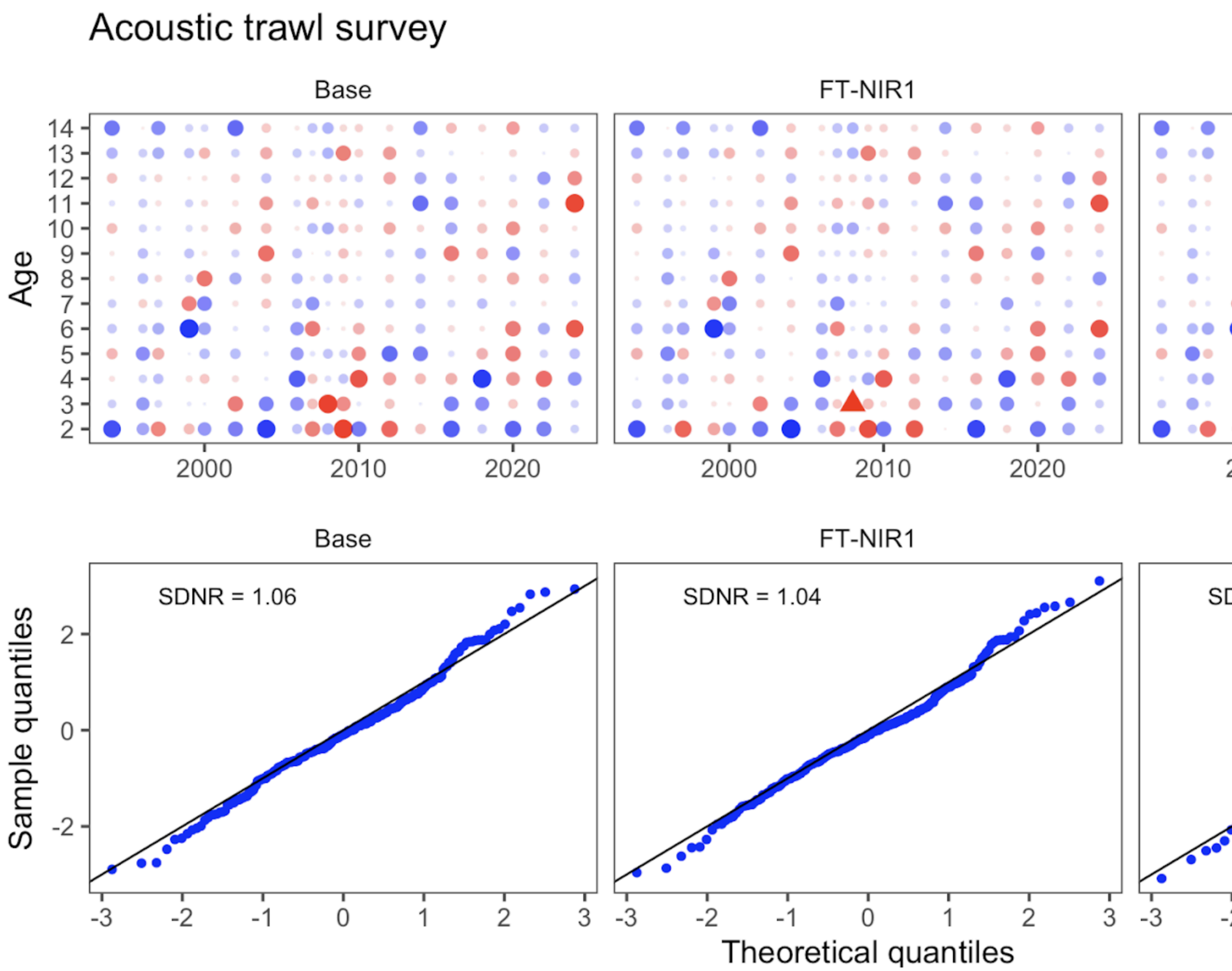

Figure 15: One-step ahead residuals for different model specifications for the acoustic-trawl survey composition data

Table 3: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	Base	FT-NIR fleets aggregated	FT-NIR fleet-specific
B_{2025}	2,800	2,500	2,700
$CV_{B_{2025}}$	0.14	0.14	0.14
B_{MSY}	1,667	1,652	1,670
$CV_{B_{MSY}}$	0.23	0.24	0.23
B_{2025}/B_{MSY}	165%	153%	162%
B_0	4,318	4,360	4,436
$B_{35\%}$	1,935	1,898	1,931
SPR rate at F_{MSY}	31%	30%	29%
Steepness	0.63	0.64	0.64
Est. $B_{2024}/B_{2024,nofishing}$	0.65	0.62	0.64
B_{2024}/B_{MSY}	182%	169%	178%

Leave-one-out survey sensitivity analysis

Here we present model runs that downweight one or more sources of survey data at a time. Table 1 shows the combinations of data types included for this analysis, which includes six model runs: (1) downweight all Acoustic Trawl Survey (ATS) data (age 1 survey index, age 2+ survey biomass index, and age composition data), (2) downweight the ATS age composition data, (3) downweight the Acoustic Vessels of Opportunity (AVO) data (survey biomass index), (4) downweight all Eastern Bering Sea Trawl Survey (BTS) data (age 1+ survey biomass index and age composition data), (5) downweight the BTS age composition data, and (6) downweight all survey biomass indices (BTS, ATS, and AVO survey biomass indices). The ATS age 1 data are included as a separate survey index in the assessment model, while BTS age 1 data are included with the age composition data in the multinomial objective function component.

Among these models, downweighting the BTS data (both solely downweighting the BTS age composition data and downweighting all BTS data) led to changes in trends over time in estimated spawning biomass Figure 16 . The model estimates of recruitment were variable across sensitivity runs, but estimates differed the most from the 2024 accepted model when all BTS data were downweighted Figure 17. Estimates of BTS survey biomass fit the BTS data very poorly in most years when the BTS data were downweighted (Figure 18). Additionally, the fits to the BTS survey biomass were affected by downweighting the BTS age composition data (without downweighting the BTS survey index). The model was relatively insensitive to other data being excluded, and fits to the ATS and AVO indices were relatively insensitive to downweighting the BTS data (see Figure 18, Figure 19, and Figure 20). The root mean square error (RMSE) in Table 2 for the model run that downweighted all BTS data was large and the and effective sample size (Eff. N) was low as compared to the 2024 accepted model.

Table 1: Configuration of different model options testing the impact of different data components

	Model description				
Dataset	Drop all BTS	Drop all ATS	Drop all indices	Drop AVO	Drop BTS Age composition	Drop ATS Ag
BTS age 2+ Index	X		X			
ATS age 2+ Index		X	X			
ATS age 1 Index		X	X			
AVO Index			X			
BTS comps	X				X	
ATS comps		X				X

Additionally, downweighting solely the BTS age composition data led to low effective sample size for the BTS.

The importance of the survey data can be seen in some of the reference calculations where the uncertainty increases dramatically and the stock-status perception was quite different across sensitivity runs Table 3. Most dramatically, dropping all survey indices led to a CV in "B_{MSY}" estimates of 1.26 as compared to 0.13 in the 2024 accepted model.

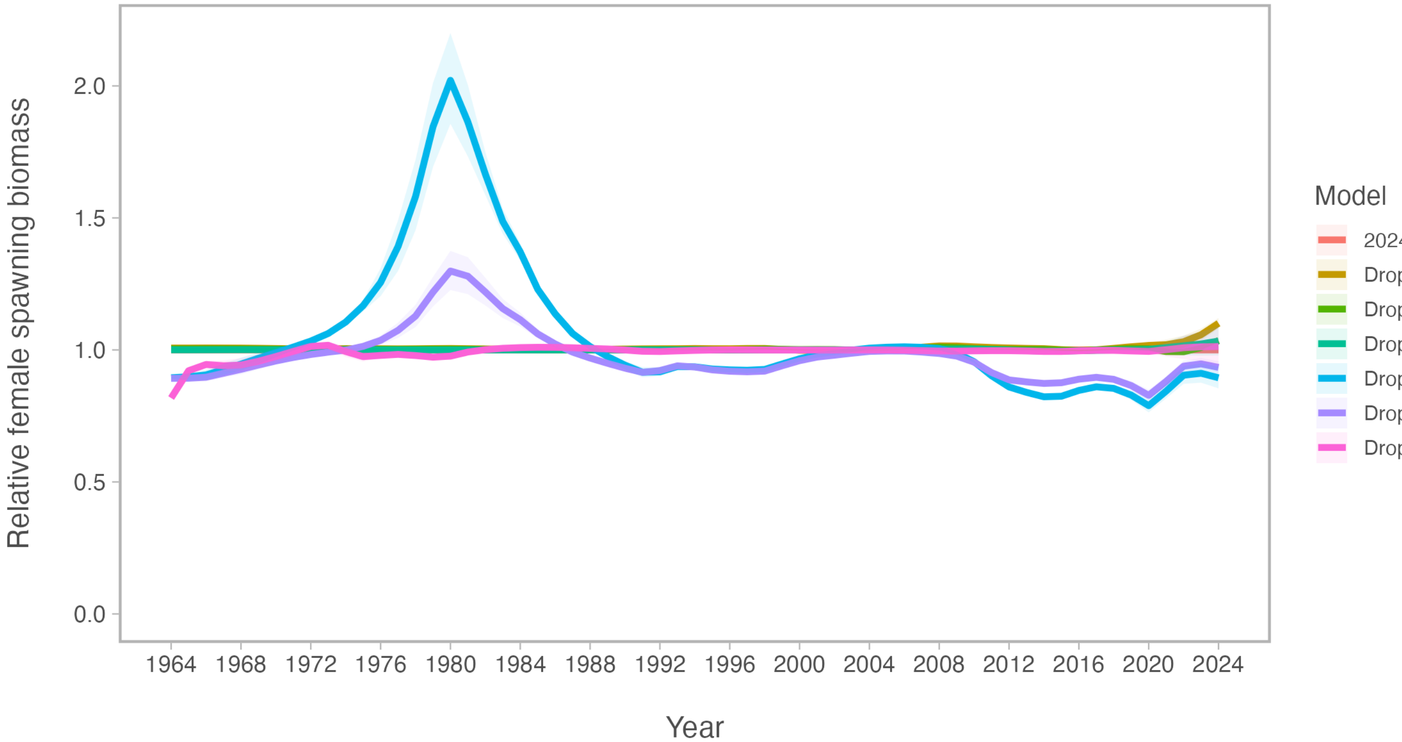

Figure 16: Comparison of spawning biomass (SSB) relative to the 2024 accepted model for models that downweight one data source at a time

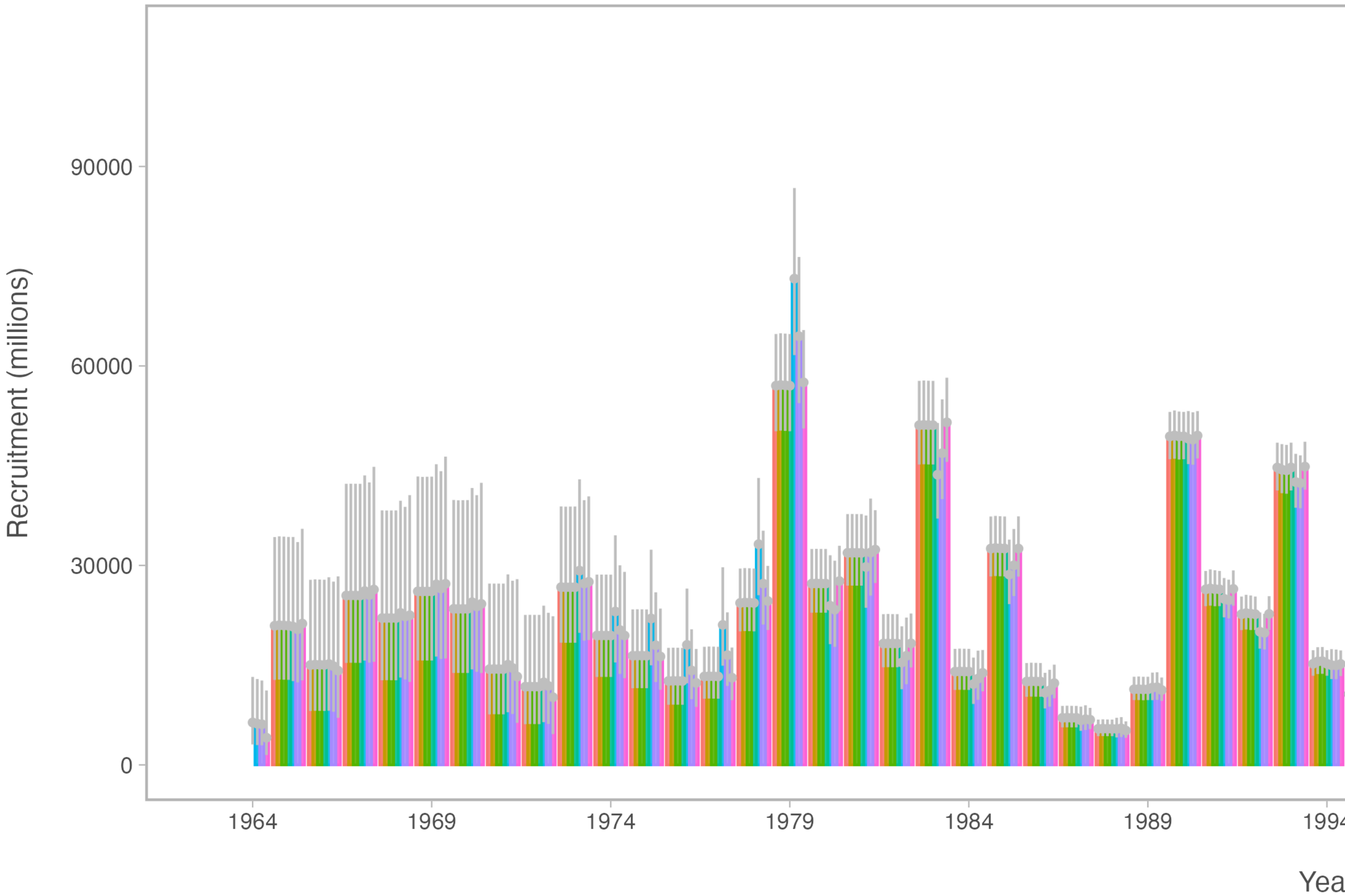

Figure 17: Comparison of recruitment estimates for models that downweight one data source at a time

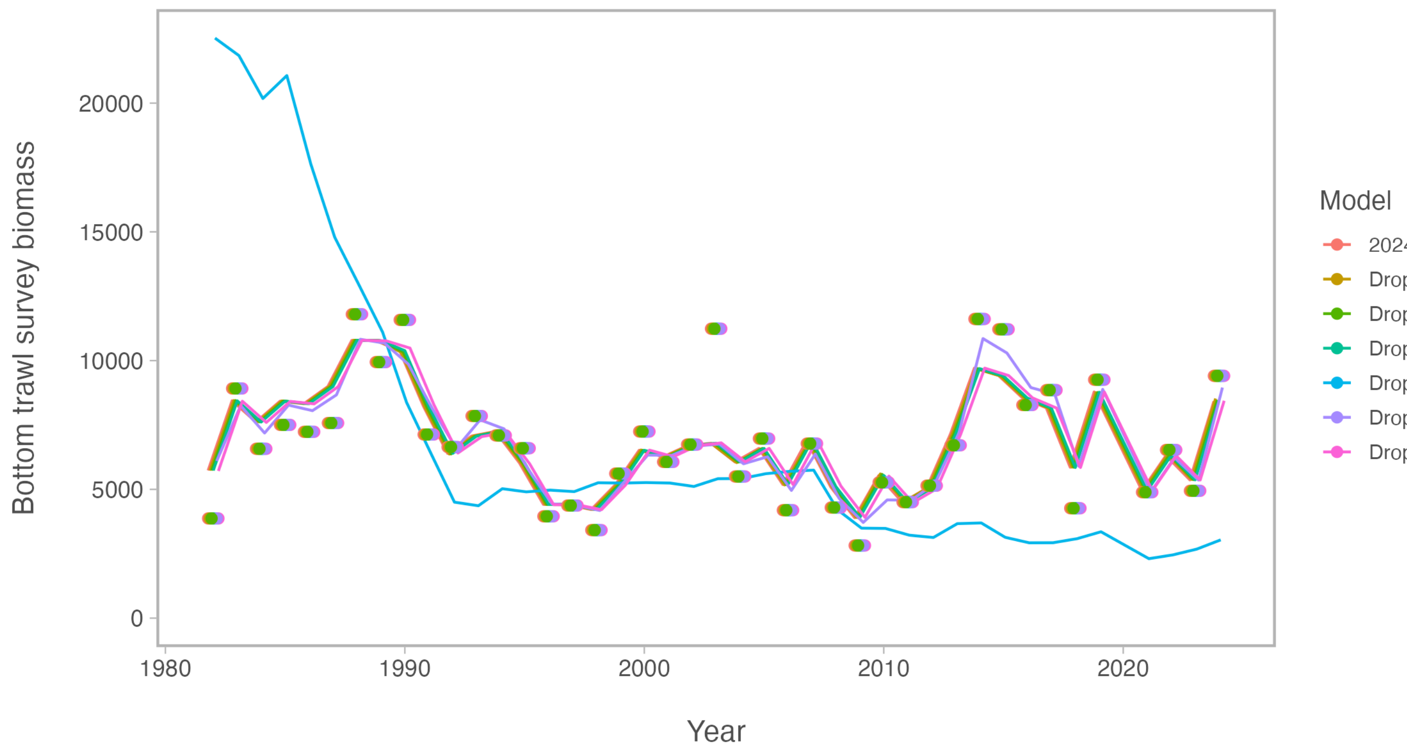

Figure 18: Comparison of fits to the Eastern Bering Sea Bottom Trawl Survey (BTS) biomass index for models that downweight one data source at a time

Table 2: Goodness-of-fit measures to primary data for leave-one-out model sensitivity runs. RMSE=root-mean square log errors, Eff. N=effective sample size for composition data).

Component	2024 accepted	Drop BTS	Drop ATS	Drop AVO	Drop indices	Drop BTS age
RMSE BTS	0.157	0.689	0.157	0.157	0.157	0.152
RMSE ATS	0.180	0.182	0.188	0.191	0.187	0.178
RMSE AVO	0.229	0.237	0.238	0.241	0.238	0.231
RMSE CPUE	0.093	0.090	0.093	0.093	0.098	0.092
Eff. N Fishery	1,182	1,379	1,190	1,185	1,176	1,344
Eff. N BTS	248	95	254	249	245	84
Eff. N ATS	269	230	164	269	273	228

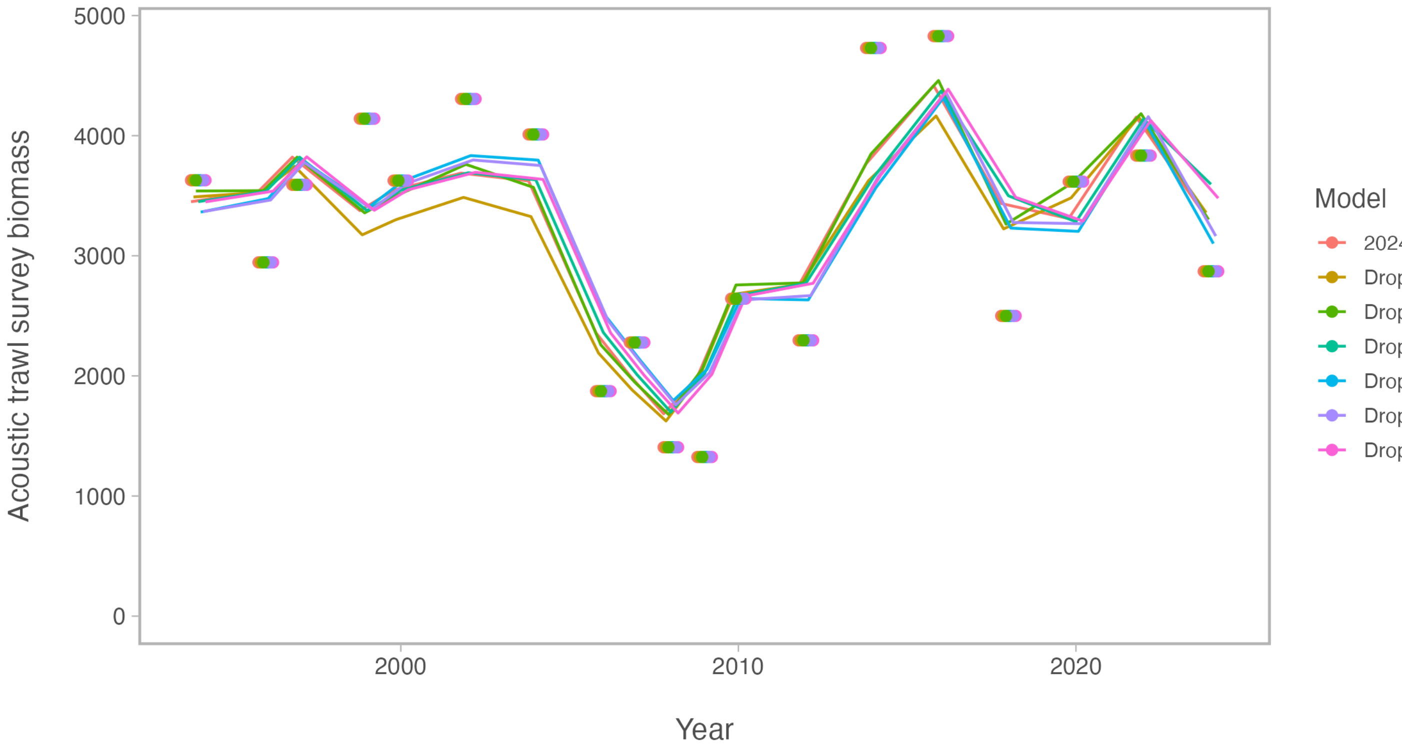

Figure 19: Comparison of fits to the Acoustic Trawl Survey (ATS) biomass index for models that downweight one data source at a time

Table 3: Estimated derived quantities and selected corresponding CVs for leave-one-out sensitivity analysis runs. See text for incremental model descriptions.

Estimated Quantity	2024 accepted	Drop BTS	Drop ATS	Drop AVO	Drop indices	Drop
B_{2025}	3,100	2,700	3,600	3,300	3,200	
$CV_{B_{2025}}$	0.13	0.16	0.14	0.16	0.18	
B_{MSY}	2,310	2,415	2,360	2,322	3,677	
$CV_{B_{MSY}}$	0.3	0.31	0.31	0.31	1.26	
B_{2025}/B_{MSY}	135%	111%	151%	141%	86%	
B_0	5,975	6,136	6,120	6,009	8,934	
$B_{35\%}$	2,066	1,988	2,101	2,080	2,074	
SPR rate at F_{MSY}	31%	35%	31%	31%	42%	
Steepness	0.62	0.59	0.62	0.62	0.52	
Est. $B_{2024}/B_{2024,nofishing}$	0.61	0.51	0.63	0.61	0.42	
B_{2024}/B_{MSY}	147%	126%	158%	151%	93%	

Figure 20: Comparison of fits to the Acoustic Vessels of Opportunity (AVO) biomass index for models that downweight one data source at a time

Incorporating multi-species M-at-age and year estimates, also total uncertainty in acoustics...

A commonly adopted approach is to use estimated natural mortality-at-age and year from multispecies trophic interaction models (e.g., Trijoulet, Fay, and Miller (2020)). The current assessment has an option to include the 2024 CEATTLE estimates of natural mortality (Table 1; Figure 21). This resulted in slightly higher recruitment (Figure 22) lower spawning biomass in the near term (Figure 23). This is due to the higher natural mortality for most ages and years compared to the base 2024 model.

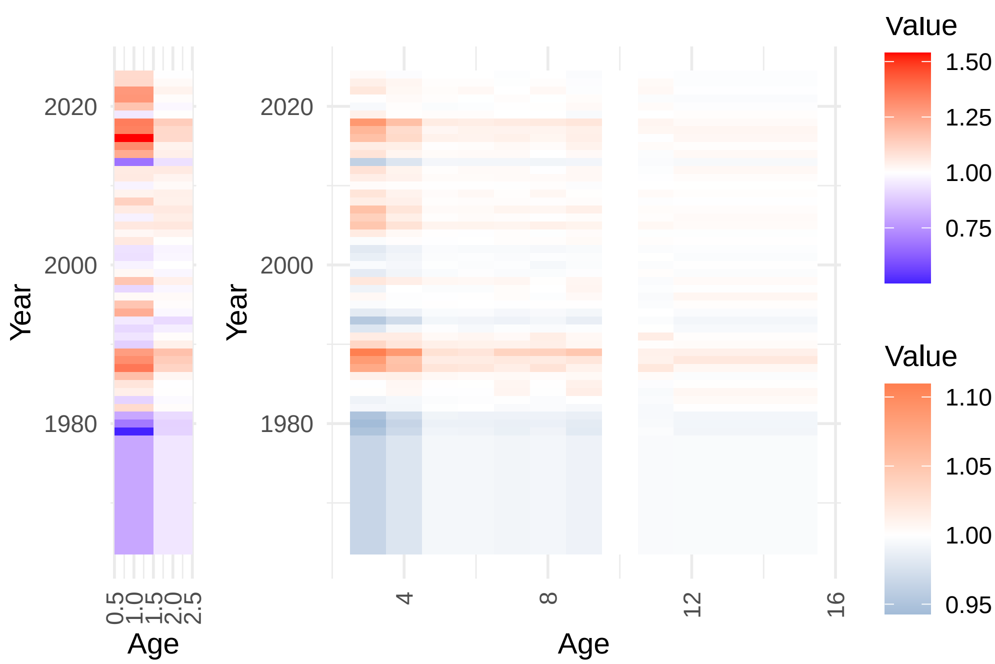

Figure 21: Depiction of the natural mortality (variability from mean value at age) as estimated from CEATTLE.

We also include a comparison of revised acoustic-trawl estimates (Urmy, Ressler, and De Robertis (2025); Figure 24). The natural mortaliity alternative run resulted in generally better fits to most data components whereas the "untuned" run with the Urmy, Ressler, and De Robertis (2025) estimates were generally poorer (Table 2). Results from the Urmy, Ressler, and De Robertis (2025) had minor impact on management-related quantities whereas those

Table 1: Natural mortality-at-age used over ages and time based on the CEATTLE 2024 model.

	1	2	3	4	5	6	7	8	9	10
1964	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1965	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1966	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1967	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1968	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1969	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1970	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1971	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1972	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1973	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1974	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1975	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1976	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1977	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1978	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1979	0.644	0.326	0.315	0.308	0.303	0.303	0.302	0.303	0.302	0.3001
1980	0.883	0.325	0.313	0.307	0.302	0.302	0.302	0.303	0.302	0.3001
1981	1.022	0.333	0.315	0.309	0.303	0.303	0.303	0.303	0.303	0.3001
1982	1.415	0.365	0.329	0.316	0.305	0.306	0.305	0.305	0.305	0.3001
1983	1.156	0.358	0.328	0.316	0.305	0.306	0.306	0.305	0.307	0.3001
1984	1.342	0.363	0.332	0.320	0.305	0.306	0.308	0.306	0.312	0.3001
1985	1.380	0.362	0.331	0.320	0.306	0.306	0.308	0.307	0.311	0.3001
1986	1.517	0.371	0.336	0.322	0.307	0.307	0.308	0.307	0.309	0.3001
1987	1.756	0.406	0.356	0.335	0.312	0.312	0.310	0.314	0.311	0.3001
1988	1.684	0.415	0.360	0.335	0.310	0.311	0.312	0.312	0.314	0.3001
1989	1.632	0.425	0.368	0.346	0.313	0.313	0.317	0.318	0.322	0.3001
1990	1.145	0.377	0.343	0.325	0.309	0.310	0.309	0.311	0.309	0.3001
1991	1.206	0.365	0.337	0.323	0.307	0.308	0.307	0.311	0.309	0.3001
1992	1.169	0.348	0.325	0.316	0.305	0.305	0.305	0.305	0.307	0.3001
1993	1.220	0.332	0.316	0.309	0.303	0.303	0.303	0.304	0.303	0.3001
1994	1.576	0.357	0.326	0.315	0.304	0.305	0.304	0.305	0.305	0.3001
1995	1.493	0.365	0.331	0.318	0.306	0.306	0.306	0.307	0.307	0.3001
1996	1.306	0.367	0.334	0.318	0.305	0.306	0.307	0.306	0.309	0.3001
1997	1.168	0.355	0.328	0.318	0.305	0.305	0.307	0.306	0.310	0.3001
1998	1.492	0.377	0.338	0.323	0.308	0.308	0.308	0.307	0.310	0.3001
1999	1.311	0.355	0.326	0.316	0.304	0.306	0.305	0.306	0.307	0.3001
2000	1.249	0.362	0.330	0.316	0.305	0.305	0.305	0.305	0.306	0.3001
2001	1.194	0.355	0.327	0.316	0.305	0.305	0.305	0.306	0.306	0.3001
2002	1.199	0.353	0.325	0.315	0.304	0.305	0.305	0.305	0.306	0.3001
2003	1.367	0.364	0.331	0.319	0.305	0.306	0.306	0.307	0.309	0.3001
2004	1.312	0.374	0.337	0.320	0.306	0.306	0.306	0.306	0.308	0.3001
2005	1.366	0.387	0.348	0.326	0.308	0.309	0.309	0.310	0.310	0.3001
2006	1.243	0.380	0.345	0.323	0.306	0.307	0.307	0.307	0.308	0.3001
2007	1.355	0.385	0.349	0.325	0.307	0.307	0.309	0.309	0.311	0.3001
2008	1.450	0.377	0.336	0.322	0.306	0.307	0.307	0.307	0.308	0.3001
2009	1.328	0.378	0.339	0.321	0.306	0.308	0.306	0.308	0.307	0.3001
2010	1.250	0.368	0.333	0.319	0.306	0.306	0.306	0.307	0.307	0.3001

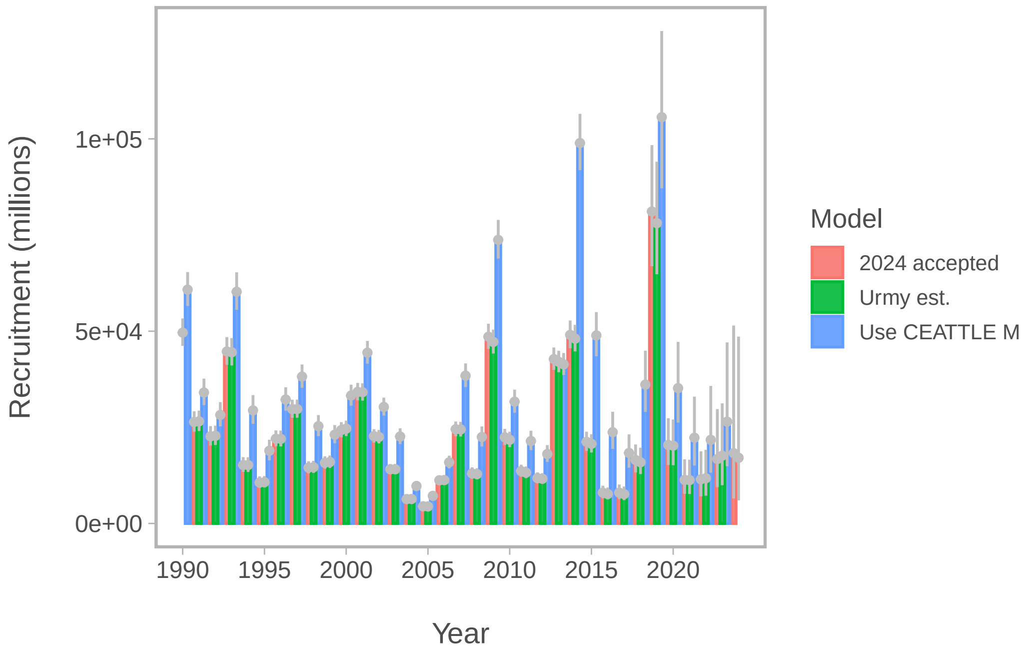

Figure 22: Comparison of recruitment estimates comparing 2024 assessment results with one where M-at-age and year is specified from CEATTLE.

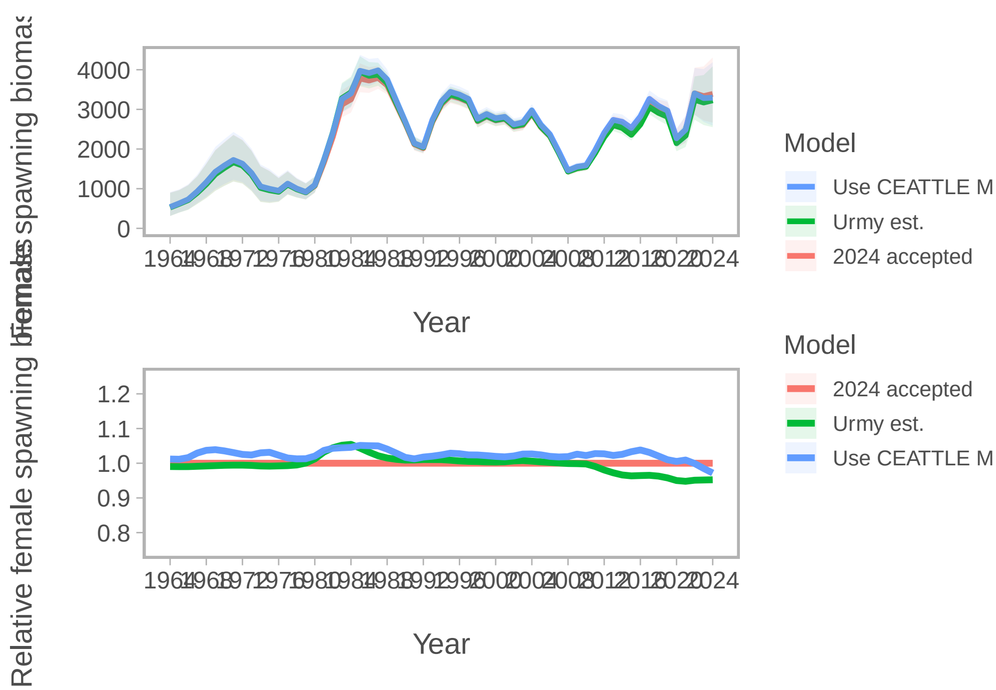

Figure 23: Comparison of spawning biomass (SSB) and SSB relative to the 2024 accepted model for model that uses the M-at-age matrix from CEATTLE.

with the alternative natural mortality specification differed in a number of ways (Table 3). The most notable difference was the estimated B_{MSY} and $B_{35\%}$ values.

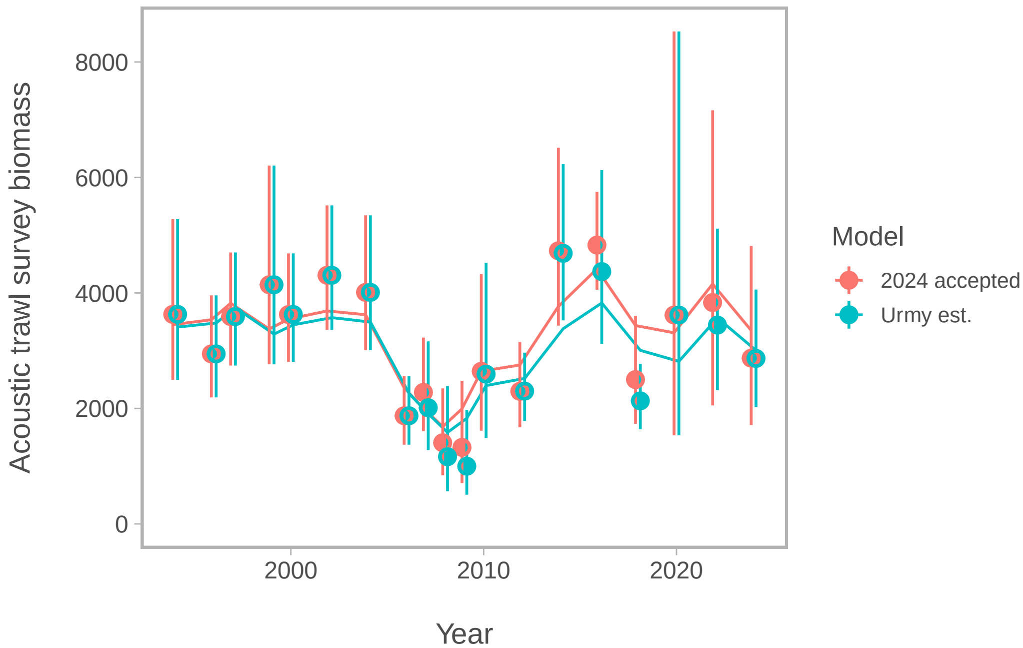

Figure 24: Comparison of fits to the Eastern Bering Sea acoustic-Trawl Survey (ATS) biomass index for the accepted 2024 model and one with revised estimates of index values and uncertainty from Urmy et al., (in prep).

Table 2: Goodness-of-fit measures to primary data for model sensitivity to applying the matrix of M-at-age and year and acoustic uncertainty estimates. RMSE=root-mean square log errors, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data).

Component	2024 accepted	Use CEATTLE M	Urmy est.
RMSE BTS	0.157	0.157	0.161
RMSE ATS	0.180	0.179	0.226
RMSE AVO	0.229	0.230	0.236
RMSE CPUE	0.093	0.093	0.093
SDNR BTS	0.950	0.960	0.970
SDNR ATS	1.000	0.990	1.220
SDNR AVO	0.980	0.960	1.020
Eff. N Fishery	1,182	1,176	1,188
Eff. N BTS	248	241	221
Eff. N ATS	269	267	278
Catch NLL	3	3	3
BTS NLL	31	31	30
ATS NLL	9	9	13
ATS Age1 NLL	11	12	14
AVO NLL	9	8	9
Fish Age NLL	167	168	162
BTS Age NLL	194	196	205
ATS Age NLL	30	31	33
NLL selectivity	196	196	199
NLL Priors	20	20	20
Data NLL	465	468	480
Total NLL	718	713	735

Table 3: Estimated derived quantities and selected corresponding CVs for sensitivity to natural mortality specification and acoustic uncertainty estimates.

Estimated Quantity	2024 accepted	Use CEATTLE M	Urmy est.
B_{2025}	3,100	3,000	3,000
$CV_{B_{2025}}$	0.13	0.13	0.13
B_{MSY}	2,310	2,584	2,290
$CV_{B_{MSY}}$	0.3	0.35	0.3
B_{2025}/B_{MSY}	135%	116%	131%
B_0	5,975	6,568	5,929
$B_{35\%}$	2,066	1,844	2,057
SPR rate at F_{MSY}	31%	35%	31%
Steepness	0.62	0.59	0.62
Est. $B_{2024}/B_{2024,nofishing}$	0.61	0.54	0.59
B_{2024}/B_{MSY}	147%	127%	141%

References

Artyukhin, Yuri B., Dmitri V. Korobov, and Yuri N. Gluschenko. 2023. "Incidental Mortality of Seabirds in Pollock Trawl Fishery in the Northwestern Bering Sea." *Trudy VNIRO* 193: 174–89. https://doi.org/10.36038/2307-3497-2023-193-174-189.

Bulatov, O. A., and D. A. Vasilev. 2023. "Assessment of Pollock Fisheries: Precautionary Approach or Maximum Sustainable Yield?" *Problems of Fisheries* 24 (3): 7–20.

Burdin, A. M. 2021. "Assessment of Impacts of the Pollock Fishery on Steller Sea Lions and Marine Mammals in the Western Bering Sea." Kamchatka Branch, Pacific Institute of Geography, Russian Academy of Sciences.

Helser, Thomas E., Irina Benson, Jason Erickson, Jordan Healy, Craig Kastelle, and Jonathan A. Short. 2019. "A Transformative Approach to Ageing Fish Otoliths Using Fourier Transform Near Infrared Spectroscopy: A Case Study of Eastern Bering Sea Walleye Pollock (*Gadus Chalcogrammus*)." *Canadian Journal of Fisheries and Aquatic Sciences* 76 (5): 780–89. https://doi.org/10.1139/cjfas-2018-0112.

Ianelli, Jim. 2025. *Ebswp: Facilitates Model Running for EBS Pollock*. http://afsc-assessments.github.io/EBSpollock/.

Japp, DW, and A Sharov. 2023. "Western Bering Sea Pollock — Surveillance Audit Report (SA1)." UCSL United Certification Systems Limited.

———. 2024. "Western Bering Sea Pollock — 2nd Surveillance Audit Report (SA2)." UCSL United Certification Systems Limited.

Punt, A. E., D. C. Smith, K. KrusicGolub, and S. Robertson. 2008. "Quantifying Age-Reading Error for Use in Fisheries Stock Assessments, with Application to Species in Australia's Southern and Eastern Scalefish and Shark Fishery." *Can. J. Fish. Aquat. Sci.* 65: 1991–2005.

Trijoulet, Vanessa, Gavin Fay, and Timothy J. Miller. 2020. "Performance of a State-Space Multispecies Model: What Are the Consequences of Ignoring Predation and Process Errors in Stock Assessments?" *Journal of Applied Ecology* 57 (1): 121–35. https://doi.org/https://doi.org/10.1111/1365-2664.13515.

Urmy, Samuel S., Patrick Ressler, and Alex De Robertis. 2025. "Estimating Uncertainty from Multiple Sources in Acoustic-Trawl Surveys." *Unpublished Manuscript*, May.